

Public Switched Telephone Network(PSTN)

From the first commercial telephones in 1876 to today's global communication systems, telephone networks have undergone remarkable transformation.

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Early Telephone Commercialization

1876: The Beginning

Telephone commercialization began in 1876. Early telephones were wired in pairs, connecting only two locations directly. There were no network connections, so users could not call more than one location.



Growth of Telephone Networks



Expansion in the 1900s

As more people adopted telephones, networks expanded rapidly.



Analog Signaling

Older networks used analog signaling, forming the basis for the PSTN.



Digitization & Automation

By the 1960s, providers digitized voice calls and replaced manual switching with automated electronic systems.



What Is the PSTN?

Definition

PSTN stands for Public Switched Telephone Network, a worldwide collection of interconnected, voice-oriented public telephone networks.

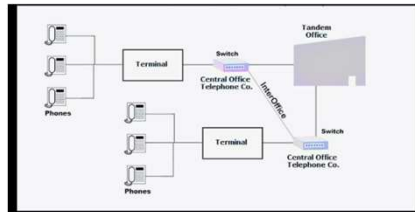
Technology

Uses traditional circuit-switched networks to connect calls globally.

Evolution

Transitioned from analog to digital signaling for improved efficiency and quality.

PSTN Infrastructure Components



Telephone Wires

Traditional copper lines for voice transmission.



Fiber Optic Cables

High-speed data and voice connections.



Switching Centers

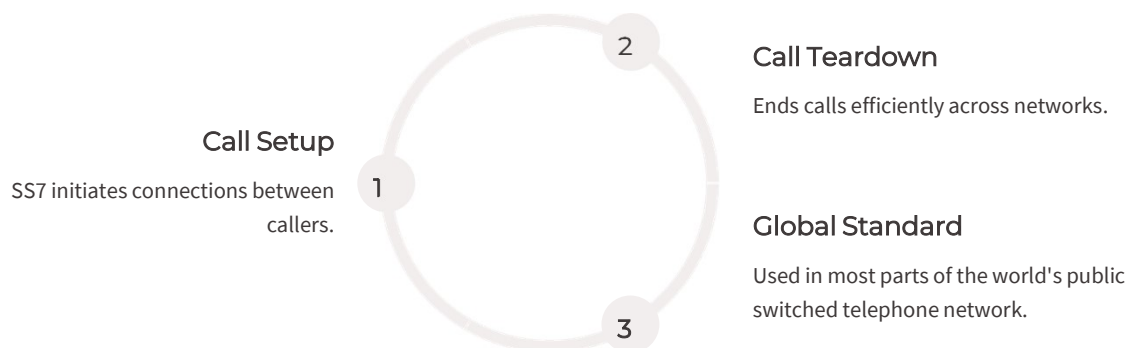
Route calls between networks.



Satellites & Cable systems

Enable long-distance and mobile communication.

PSTN Signaling: The SS7 Protocol



Advantages of PSTN

1

Reliable During Outages

PSTN works even when the power is out.

2

Clear Call Quality

Provides consistent and clear voice communication.

3

Low Delay & Drop Rate

Less chance of delays or dropped calls compared to some alternatives.

Disadvantages of PSTN

1

Higher Costs

PSTN calls, especially long-distance and international, are often more expensive than digital alternatives.

2

Limitations

PSTN offers fewer advanced feature compared to VOIP. Things like video-calling or voice email-to-email are not available